Prediction of Song Popularity Using Machine Learning Concepts

Adit V. Kaneria, Abishek B. Rao, Shivani G. Aithal, and Smitha N. Pai

1 Introduction

The music companies invest a huge amount in search of song which can make an impact. An ideation on what makes a given song to be influential can bring huge profit to the music industry. This idea can help in predicting the success of the given song before the launch of the album. Good amount of work is carried out on the lyrics of the song. Automatic predicting a song to be hit is a challenge. With digitization, large amount of information about the song can be captured, reproduced, reworked and remodeled. Musical tastes vary from person to person and in today's world, it is highly individualistic. Streaming services like Spotify have realized this and have employed algorithms to understand an individuals' music taste and recommend music based on their listening history. This helps the user in getting a personalized playlist. However, as things are getting easier for consumers, artists have a tough time churning out good music every now and then. A hit song can do wonders for their careers and launch them straight into the spotlight. There is a lot of uncertainty when they launch a new song as to whether it will be accepted by consumers worldwide. It is a heavy investment for producers too. With the advent of data science and machine learning, industries have changed their approach to business. The availability of data has given them an opportunity to predict outcomes more accurately and with greater certainty. The music industry can also apply machine learning algorithms to see if a song will be a hit or no. This way they can make minor tweaks, keeping the artists music vision in mind, and appeal to a greater audience. The music industry around the globe consists of companies and individuals that make huge money by creating songs and selling them to the people by promoting their songs in live concert, shows, etc. There is no guarantee that a signal in the song might become popular resulting

A. V. Kaneria · A. B. Rao · S. G. Aithal · S. N. Pai (✉)
Department of ICT, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India
e-mail: smitha.pai@manipal.edu

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2021

S. K V and K. Guruprasad Rao (eds.), *Smart Sensors Measurements and Instrumentation*, Lecture Notes in Electrical Engineering 750,
https://doi.org/10.1007/978-981-16-0336-5_4

in high profit while some signal may turn out be not so popular which may lead to losses. In the project, analytics is applied to predict if a song can be classified as belonging to the top 100 of the official chart so that it can be classified as a hit song. Popularity of a track can be affected by other aspect like the social, economic, etc.

The prediction of song popularity can be used in various applications. Some of them are as follows: As a gesture of recognition, playing the popular song of the famous personalities as they arrive randomly at a gathering on recognizing their face. Recognizing the emotional expression of the person and play music as per their mood. Guessing the age range of the person and play music according to their generation. Popular music based on their native region can also be played on recognizing the person's place of origin. Music can also be played as per the situation, sick, patriotic, gaming, etc. to enhance their mood.

The work currently focuses on the features of the audio of the music track.

2 Related Work

Machine learning uses various algorithms for the purpose of predictions. Good amount of work is carried out to predict the popularity of a song. A study by Stanford students on the problem "Hit Song Science" [1] used dataset of 4000 songs and extracted features of the song from Spotify API. Among the different algorithms used, the best results were obtained using neural network with one hidden layer and logistic regression. It provided an accuracy up to 75%. Another study [2] in the same topic used a database of songs collected between the years 1985 and 2013. Music features of the songs were extracted from the Echo Nest [3], the world's leading music analytics organization with over a trillion data points on millions of popular songs. Five classifier models are used. All the models are tested with and without feature selection. Validation process of tenfold is used for verification on the models. The paper addressed that hit songs can be classified as hit or non-hit using audio features without any metadata. The authors had built an application, where users can fill in audio features of a new song and classify it as a hit or not a hit. Another study [4–6] tried to determine the change in popularity of songs over time to predict how popular songs in the future will turn out to be. The paper takes attributes of popular songs from the 1900s to today and analyzes how the attributes contribute to the song's genre and year of release. Study makes a comparison of hit songs from UK and China. Features for the track are obtained from the work [3]. The work [7] uses lyrics and acoustic-based features. Dataset contained songs by prominent artists from different generation. During the study, the range of number of tracks in the chart in the selection of hit and non-hit is taken differently by each of the author. Author [8] uses wide and deep neural network for and uses the features of audio that are low and high levels. This process is used is used to analyze the hit songs. The work [9] emphasizes the various moods revolving each of the songs. Songs could be of dancebability, rock, relaxed songs and the various moods party-like, electronic, etc. associated with it is addressed. In the work [10], two models support vector machine

(SVM) and long short-term memory (LSTM) are used to find the popularity of the artist. Many features are used as the input, and if the features are too fine grained, then the results could be counterproductive. Results showed that performance of LSTM is better compared to the LSTM. Deep learning technique using the Mahalanobis distance vector scores is used to find the next song which need to be played in the playlist which is recommended to the user. The work [11] uses the feature extraction from the lyrics, deep neural network autoencodes to carry out the decision for popular songs. In the work [12], the authors have found that there is difference in readability of the lyrics and its repetitiveness between the genre. The top artists lyrics are repetitive but may not be simple when it comes to readability. The author [13] discusses the tools and techniques used in developing context-aware services. Context-awareness involves cognitive psychology, perception of music by human, social computing and recognition of emotion in music. The author [14] uses the wide and deep neural network which exploits the high- and low-level audio features to predict hit songs. The study [15] revealed that it was easy to predict Chinese hit song more easily than the UK. This suggested that the characteristic for hit song varied with region of residence. Linear regression and support vector machine (SVM) were used for comparison. Hit songs were taken from upper half of the top chart and non-hit the lower half from the non-hit list. Using SVM and classifiers, they distinguished the top 1% of the hit song from others with promising result. Authors have also used other matrix to measure the popularity like the kurtosis, debut, standard length, mean, maximum skewness and sum [16]. A relation between artist and song is brought about in the work [17]. LSTM and SVM are used for comparison, and SVM is proved to be better.

Emotional well-being of a person is vital for the cognitive process. It has its influence on the learning, memory and other factors. Emotions can be captured using various methods like electroencephalography, electrocardiography, electromyography and bio-signals. Music can influence for the well-being of the individual. Happiness and sadness are the easily recognized emotions using speech recognition system [18]. There are ways of communicating emotions like usage of works, voice tone, body language, facial emotions, variations in heart rate and conductance in the skin [19]. The author addresses how emotions can be recognized using the highly discriminative features and combining the orientation and texture of the images captured. The process is effective even in the case of occlusion and illumination along with noise.

The current papers implement certain machine learning algorithms on a dataset consisting of features extracted from the Spotify database. Data preprocessing techniques were used to handle missing values and normalization. A total of four models were used–random forest, logistic regression, naïve Bayes and decision tree [20, 21]. The main work is focused on supervised learning, as the dataset has labeled output. The dataset contains hit and non-hit songs. Models should be able to recognize the pattern which can categories them along with their label.

3 Background Theory

3.1 Logistic Regression [22]

Logistic regression as the name implies computes the log of the features from the music track. The real values obtained are mapped to values between 0 and 1 as per (1) to obtain the output y.

$$y = 1/(1 + \mathrm{e}^{-x}) \tag{1}$$

x forms the sum of the combination of the weights and feature vector values. Alternatively, output y is represented as in (2) as

$$y = \mathrm{e}^{(w0 + w1 * x1)} / \left(1 + \mathrm{e}^{(w0 + w1 * x1)}\right) \tag{2}$$

w_0 is the intercept term, and w_1 is the weight associated with the feature vector $\times 1$. Similarly, a single weight value is associated with each of the column feature vector. The weight value is obtained after training the model.

3.2 Decision Tree [23]

Decision trees take bulk of data and are split in groups recursively as in a tree, until the data is classified into the required class at the leaves. It is also a supervised learning technique.

Mainly, two types of trees whose decisions are categorical and regression (continuous values).

Shannon entropy $E(S)$ measures how much of uncertainty or randomness exists in a data for a finite set S.

$$E(S) = \sum_{x \in X} p(x) \log_2 \frac{1}{p(x)} \tag{3}$$

It intimates about the predictability of a certain event. Higher uncertainty is due to higher values and lower with lesser value.

Kullback–Leibler divergence KL(S, A) (Information gain) with set S and choosing the attribute A shows the change in the entropy. It is a measure of changes in the entropy with regard to the independent variables.

$$\mathrm{KL}(S, A) = E(S) - E(S, A) \tag{4}$$

Alternatively,

$$\mathrm{KL}(S, A) = E(S) - \sum_{i=0}^{n} P(x) * E(x) \tag{5}$$

Entropy for the full set is $E(S)$ and $E(S, A)$ on applying the feature A. Probability of the event x is $E(x)$.

3.3 Naïve Bayes [24]

Naive Bayes classifier is based on the Bayes' theorem, which makes a presumption that input variables or features are not dependent on each other. Bayes' theorem, which is centered on likelihood or conditional probability, which says if an event (C) has occured, with what likelihood the event (B) would occur is given by (6).

$$P(B|C) = (P(C|B)P(B))/P(C) \tag{6}$$

P is the probability.

P (B|C) implies given with what probability will the event B occur, given that event C has occurred already.

P (C|B) implies given that B has occurred already, with what probability C is going to occur.

$P(B)$ signifies the probability of the event B occurring.

$P(C)$ signifies the probability of event C occurring.

The training time in the case of naïve Bayes is low. If the conditional dependence between the attributes holds, it will converge faster than the discriminative models like that of logistic regression. The size of the model is small.

3.4 Random Forest [25]

Random forest consists of large collection of decision tree which work like an ensemble. Each tree provides the class to which the features belong. The class with majority of vote signifies the class to which it belongs. The work suggests that a collective decision of large number of unrelated models of tree will provide a better result than a single tree. The error if introduced by individuals will get suppressed unless all of the models are providing erroneous results. The results obtain are better if models are created around specific information of the features rather than randomly selection. Results obtained could also improve if the error introduced by tree does not correlate with others. The model is large. It has to rebuild the model with quick changes in the data.

3.5 K-Nearest Neighbor (kNN) [26]

The k neighbors are chosen to form cluster. The choice of k has a large image on the results obtained from the algorithm. With large vale of k, the effect of variance because of random error is reduced but may lose out on small information which might be importance. The correct value of k brings a balance between under and over fitting.

3.6 XGBoost [27]

This algorithm used gradient boosted decision tree for increasing the speed and performance. New models are added to existing model in order to reduce the errors introduced by them. It uses the gradient descent algorithm to reduce the errors.

4 Methodology

The project uses dataset for the audio features from Spotify Web API. The following features are available [28]:

a. **Danceability**—It provides a value which signifies how much it is meant for dancing. This value is based on the elements of music like beat strength, tempo, overall regularity and rhythm stability. Zero value signifies low hit, and value of one indicates high hit for the dance.
b. **Energy Level**—Measures how intense the activity is. This can be related to loud, noisy and fast track. The overall value is based on features like timbre, dynamic range, onset rate, loudness and general entropy. The values can be one or zero based on the liking or not.
c. **Key**—Specifies the track the key is in. Corresponding integer values are suitably chosen.
d. **Loudness**—Average loudness of the track is measured in decibels (dB). Values are between the range −60 and 0 dB.
e. **Mode**—The scale from which the melodic content is obtained. Values are one and zero for major and minor values.
f. **Speechiness**—It identifies if there exist any spoken words in the track. A value below 0.33 represents only music without the speech content. Between 0.33 and 0.66 signifies both music and speech are present in sections. A value of 0.66 to 1 indicates that the track is entirely made of spoken words.
g. **Acoustisness**—It gives a measure between 0 and 1 about the behavior of sound waves (acoustic) of the track.
h. **Liveness**—Detects for vocal content. Value close to 1 depicts that the track has no vocal content. Value above 0.5 is for tracks with instrumental content.

i. **Valence**—Cheerfulness and positivity are indicated by this value. High value of 1 represents positivity and 0 with sad and other negative aspects.
j. **Tempo**—It is the estimated average beats in the track for every minute. It also suggests the pace or speed of the song.
k. **Duration**—The length of the track in milliseconds.
l. **Time Signature**—Specifies how many beats are present in each measure.
m. **Instrumentalness**—This includes some vocal words. Closer is this value to the 1, implies it does not have the vocal words.

Some other features which are present in the dataset like genre, artisit_name, track_name and track_id are not used for predicting popularity. There is a feature called "popularity" which is used as the true output. This results obtained by using various models are compared with this feature which is available. The machine learning work flow is separated in five parts: extraction of data, preprocessing the data, building the model, evaluation with the test set and prediction the results [29]. Following is the steps carried out in the current work.

4.1 Extract Data

Spotipy extracts music features from the Spotify database and stores them in a csv file. The API takes the artists names as input and returns the following attributes—acousticness, danceability, loudness, tempo, speechiness, liveness and so on. The API will require a developer account to be created with Spotify to access their database. On logging the execution of the content at the given URL is proceeded. The attributes are fetched and stored into the destination csv file. The dependent variable values are obtained by scraping the data from an official chart website. For a song to be a hit, it must have reached the top 100 in the official chart.

4.2 Preprocess Data

During data preprocessing, null values have to be removed. This is especially in the case of random forest. Datasets with only metadata are chosen. Audio features within the specified range are selected. Duplicates in the dataset were avoided. Dataset is split into 80 percent training and 20 percent testing set with both the set containing equal proportion of hit and non-hit songs.

4.3 Model Building

Machine learning model, code is written in Python (v3.6) in sublime text editor. It is an open-source text editor that can implement data visualization, statistical modeling, machine learning, etc. The machine learning part of the code is in Python, and the library ScikitLearn is used.

Four different algorithms decision trees, logistic regression, support vector machine and random forest are used. Both linear and nonlinear models are used. The seed is fixed to have the same split in the data with each run. This enables uniformity and accuracy while the results are compared across different models. Models are built to classify the song into hit and non-hit using the rules that are obtained on training with the various algorithms.

4.4 Evaluation of the models

The models obtained on training have to be cross-checked for expected results. To incorporate this, the dataset is divided into train and test set. There are various validation techniques, of which, K-fold cross validation is one of them. The dataset is partitioned into K subset each of same size. Each of these is referred to as fold. Algorithms are trained on the K-1 folds and tested on the remaining one fold. This process is done K times each time with a different partition as test and train. Average of this is used for future prediction. Currently, a tenfold validation is used for estimating the accuracy. Accuracy refers to the ratio of correctly identified class to the total number of instances in the correct class. All the four models in the current project use the tenfold cross-validation process.

Confusion matrices are useful in finding the accuracy. The information provided in the matrix is recall, precision, f1-score and the support for each model. The four region of the matrix stores information about the predicted outcome and actual outcome. Figure 1 shows the confusion matrix with the false negative (FN), true positive (TP), true negative (TN) and false positive (FP) values. The actual explanation with respect to hit and non-hit is also provided.

The metrics of evaluation of the model like f1-score, accuracy, recall and support can be obtained from the confusion matrix.

		Predicted class	
		P	N
Actual Class	P	**True Positive(TP)** Songs that are actually hit and rightly predicted by the model.	**False Negative (FN)** Songs predicted as non-hit but are actually a hit.
	N	**False Positive (FP)** Songs that are actually not a hit but predicted as hit by the model.	**True Negative(TN)** Songs that are not a hit actually and are rightly predicted by the model.

Fig. 1 Confusion matrix

The **precision** of the model specifies how correctly it is able to not label a non-hit song as a hit song. It is the ratio of TP to (TP + FP) from values of confusion matrix.

Recall of the model specifies how correctly it is able to find all the hit songs. It is the ration of TP to (TP + FN).

Accuracy suggests what percent of the prediction carried out by the model were correct. It is given by the ration of (TP + TN) to (TP + TN + FP + FN). Accuracy is the most preferred metric to evaluate a model.

F1-score provides a balance between the precision and recall. F1- score or measure $= 2 \times ((\text{precision} \times \text{recall})/(\text{precision} + \text{recall}))$. It is the harmonic mean of precision and recall.

Area under curve is used to check how capable the model is, to distinguish between the classes.

4.5 Extension of Project for Related Application

Recognizing of face process involves creation of individual folder with their names. Within each person's folder, various subfolders are created one each for each facial emotion. The emotions that are currently recognized are surprise, fear, anger, happiness, sadness, disgust and neutral. For illustration purpose, Japanese Female Facial Expression (JAFFE) [30, 31] database is used in the implementation. During implementation, live feed of faces are carried out. But results in the paper are confined to only JAFFE database due to the restriction to reveal any body part. The result shows 96% accuracy on the still images. The actual implementation has people from other country originals as well. The accuracy of the result could reduce if the subject is on the move and not directly facing the camera. The face region of interest is obtained using the multi-task cascaded neural network (MTCNN) [32, 33]. Figure 2 shows the CNN model used in the implementation of the work.

The face recognition is carried out using the classifier and is carried out using Fisher discriminant analyzer [34]. Song pertaining to a particular region can be played on recognizing the face contour. Figure 3 shows the high level diagram representing the flow of events in the project. Classifying the song based on emotion is a part which

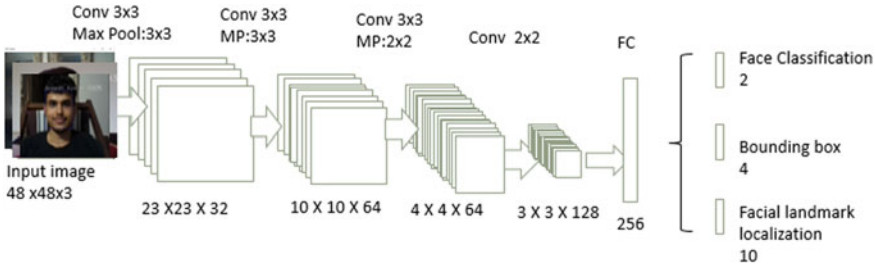

Fig. 2 Facial recognition using MTCNN[28]

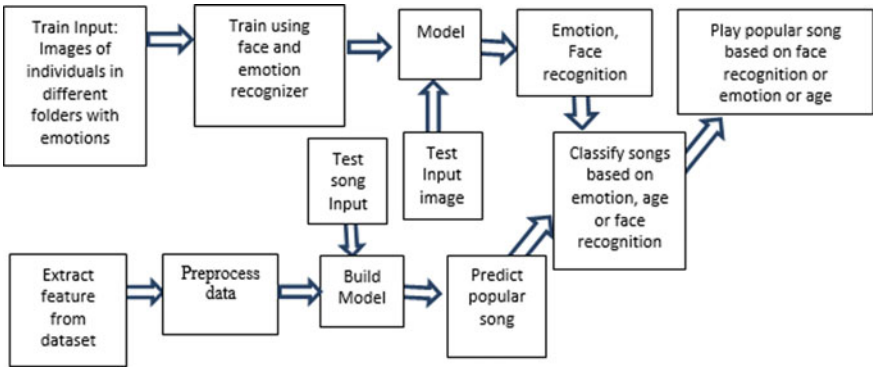

Fig. 3 Overall work flow of the project

is proposed and not yet completed. The emotion recognition is carried out using the trained model mini-Xception. [35]. Songs for different age group can be played as soon as face is recognized and age is computed. Age and gender can be computed using the CNN model [36]. It has three layers with 96 nodes in the first layer with kernel size 7, second layer with 256 nodes with kernel size 5 and third layer of 384 nodes with kernel size 3. The photos available in the dataset are 26580 with number of subjects as 2,284 annotated with age and gender. The age is grouped into groups of (0–2, 4–6, 8–13, 15–20, 25–32, 38–43, 48–53, 60-). The younger group has their facial features changing faster, so smaller size variation in age is used. The problem is changed from regression to classification by using this grouping.

5 Results and Discussion

A correlation between every feature with every other feature is analyzed in Fig. 4. Energy and loudness have a strong correlation. From the values that are visible, popularity is better correlated with loudness, followed by danceability, energy, valence, tempo and duration. Popularity is negatively correlated with to acoustics, speechiness, liveness and instrumentalnes.

The experiments were carried out for all the four models using the tenfold validation. The results of the prediction are shown in Table 1 in the form of a confusion matrix. The result shows that random forest with highest value of true positive and true negative among all other models is the best algorithm. The size of the dataset is 37236 songs.

Table 2 depicts the precision, recall, f1-score, accuracy and AUC statistic of the various models. Most promising among this is random forest with an accuracy of 89%.

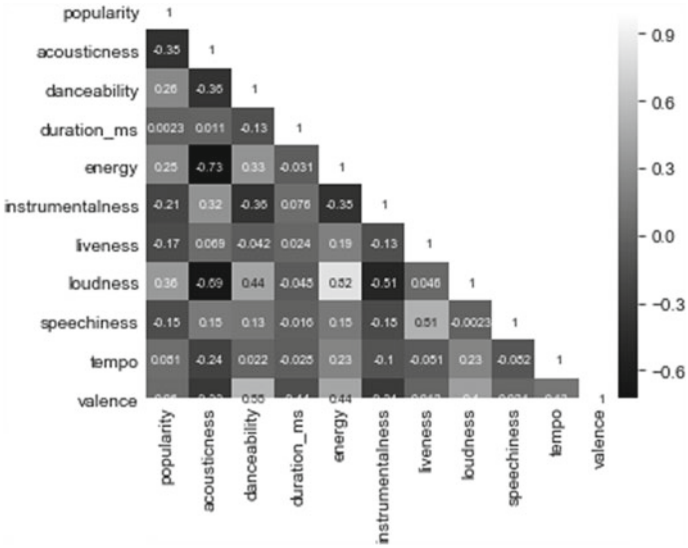

Fig. 4 Correlation between various features

Table 1 Confusion matrix using various models

Predicted class				
	Model		Hit	Non-hit
Actual class	Decision tree	Hit	23,869	4097
		Non-hit	2361	6909
	Logistic regression	Hit	27,349	617
		Non-hit	8495	775
	Naïve Bayes	Hit	14,363	13,603
		Non-hit	1997	7273
	Random forest	Hit	26,739	1227
		Non-hit	3015	6255
	K-nearest neighbor	Hit	23,842	4124
		Non-hit	4909	4361
	XGBoost	Hit	27,344	622
		Non-hit	8366	904

The data distribution is such that random forest provides highest accuracy. Decision tree also works on a similar concept, hence provides good accuracy and is comparable with random forest. Naïve Bayes works with the assumption that features are independent of each other. The popularity of music is dependent on many features in the song as seen from the correlation matrix. So, naïve Bayes cannot be used for this classification problem. It has the lowest accuracy. KNN presumes that the

Table 2 Classification report of various models

Model		Precision	Recall	$F1$-score	Accuracy	AUC
Decision tree	Hit	0.93	0.88	0.90	0.85	0.81
	Non-hit	0.62	0.74	0.68		
Logistic regression	Hit	0.79	0.99	0.88	0.79	0.51
	Non-hit	0.55	0.03	0.05		
Naïve Bayes	Hit	0.90	0.52	0.66	0.85	0.64
	Non-hit	0.30	0.78	0.43		
Random forest	Hit	0.91	0.97	0.94	0.91	0.81
	Non-hit	0.87	0.66	0.75		
KNN	Hit	0.85	0.88	0.87	0.79	0.65
	Non-hit	0.49	0.41	0.45		
XGBoost	Hit	0.79	0.99	0.88	0.79	0.51
	Non-hit	0.56	0.03	0.05		

neighboring data within the same cluster/class have similar characteristic. There are some features which are not correlated with each other, and hence, nearest neighbor do not provide as good result as the decision tree algorithm. Logistic regression is a simple classifier which presumes there is a linearity between dependent and independent variable which is not true in this example, and hence, it does not provide the required accuracy. Though XGBoost has provided the good accuracy, it has low false negatives.

6 Conclusion

The study showed that it is possible to predict hit songs with reasonable accuracy. Inclusion of metadata could enhance the accuracy of the results. Inclusion of additional features could provide improvement on the results. Some of them include the popularity of the artist, the lyrics, etc. The inclusion of age factor helps in predicting the song desired by that age group. Identification of face using the face recognition technique helps in playing the song of the popular artist. The songs can be played based on the emotional state of the individual. This can help the sick or emotionally depressed people to recover at a faster rate. A relative comparison of various protocols to compare the popularity of the result is carried out using decision tree, random forest, naïve Bayes, logistic regression, KNN and XGBoost. Random forest provides good results as compared to other algorithm in terms of accuracy. The distribution of data is better suited for decision rule and hence works well both with random forest and decision tree.

7 Future Scope

The study presumed some pre-assumed features for carrying out the project. Since a number of features are present in a song, the use of principal component analysis or enhanced feature extractor or selector could help in obtaining better results. Work can be substantiated with other database which includes several other features. The artist and lyrics also contribute to the popularity of the song. These features can be incorporated to make better predictions.

References

1. Francois, P.: In: Li, T Tzanetakis, G., Ogihara, M. (eds.) Music Data Mining, pp. 305–326. CRC Press (2011)
2. Braheny, J.: Craft and Business of Songwriting, 3rd edn. F & W Publications, New York (2007)
3. The EchoNest, Available: https://the.echonest.com.
4. Leikin, M.A.: How to Write a Hit Song, 5th edn. Hal Leonard (2008)
5. Perricone J.: Melody in Songwriting: Tools and Techniques for Writing Hit Songs. Berkley Press (2000)
6. Borg, N., Hokkanen, G.: What Makes For a Hit Pop Song? What Makes For a Pop Song? CS229 Report (2011)
7. Interiano, M.: Muscial trends and predictability of success in contemporary songs in and out of the top charts. J. Royal Soc **5**(5) (2018)
8. Schedl, M.: Deep learning in music recommendation system. Frontiers Frontiers Appl. Mathe. Statist. **5**(44), 1–9 (2019)
9. Yu, H., et al.: Popularity Prediction for Artists Based On User Songs Dataset, pp. 17–24. ICCAI, Indonesia, ACM (2019)
10. Tran, T., Seeney, R., Lee, K.: Adversarial Mahalanobis Distance-based Attentive Song Recommender for Automatic Playlist Continuation", SIGIR' 19, pp. 245–254. ACM, France (2019)
11. Martín-Gutiérrez, D., Hernández, P.G., Belmonte-Herández, A., Álvarez, G F.: A multimodal end-to-end deep learning architecture for music popularity prediction. IEEE Access **8**, 39361–39374 (2020)
12. Schedl, M.: Genre Differences of Song Lyrics and artist Wikis: An Analysis of Popularity, Length, Repetitiveness and Readability, pp. 3201–3207. IW3C2, ACM (2019)
13. Kaminskas, M., Ricci, F.: Contextual Music Information Retrieval and Recommendation: State of The Art and Challenges. Comput. Sci. Rev. 1–31 (2012)
14. Zangerle, E., et al.: Hit song prediction: leveraging low-high level audio features. IN: 20th ISMIR Conference, Netherlands (2019)
15. Fan, J., Casey, M.A.: Study of Chinese and UK hit songs prediction. In: Proceedings of International Symposium on Computer Music Multi-Disciplinary Research, pp. 640–652 (2013)
16. Lee, J., Lee, J.: Music popularity: metrics, characteristics, and audio-based prediction. IEEE Trans. Multimedia **20**(11), 3173–3182 (2018)
17. Yu, H., Li, Y., Zhang, S., Liang, C.: Popularity Prediction for Artists Based on User Songs Dataset, ICCAI '19, April 19–22, 2019, p. 1734. Indonesia ACM, Bali (2019)
18. Maria, E., Matthias, L., Sten, H.: Emotion recognition from physiological signal analysis: a review. Electron. Notes Theoret. Comput. Sci. **343**, 35–55 (2019)
19. Mahmood, A., Hussain, S., Iqbal, K., Elkilan, W.S. Recognition of facial expressions under varying conditions using dual-feature fusion. J. Comput. Intell. Image Process (2020)

20. Daumé, H.: III. Machine learning, A course in machine learning. CIML (2012)
21. Han, J., Kamber, M., Pei, J.: Data mining: concepts and techniques. Morgan Kaufmann, Elsevier (2016)
22. Peng, C., Kee, K.L., Ingersoll, G.M.: An introduction to logistic regression analysis and reporting. J. Educ. Res. **96**(01), 3–14 (2002)
23. Navada, A., Ansari, A.N., Patil, S., Sonkamble.: Overview of use of decision tree algorithms in machine learning. In: 2011 IEEE Control and System Graduate Research Colloquium, pp. 37–42. Shah Alam (2011). https://doi.org/10.1109/ICSGRC.2011.5991826
24. Rish, I.: An Empirical Study of the Naïve Bayes, pp. 41–46, Research Article (2001)
25. Biau, G.: Analysis of random forests model. J. Mach. Learn. Res. **13**, 1063–1095 (2012)
26. Zhang, Z.: Introduction to machine learning:k-nearest neighbor. Ann. Transl. Med. **4**(11), 1–7 (2016)
27. Chen, T., Guestrin, C.: XGBoost: A scalable tree boosting system. In: International Conference on Knowledge Discovery and Data Mining, pp. 785–794 (2016)
28. Spotify for Developers, https://developer.spotify.com/community/news/2016/03/29/audio-features-recommendations-user-taste
29. Herremans, D., David, M., Sörensen, K.: Dance hit song prediction. J. New Music Res. **43**(3), 291–302 (2014)
30. Lyons, M.J., Akamatsu, S., Kamachi, M., Gyoba, J.L Coding facial expressions with gabor wavelets. In: 3rd IEEE International Conference on Automatic Face and Gesture Recognition, pp. 200-205. The Japanese Female Facial expression (JAFFE) Dataset (1998). https://doi.org/10.5281/zenodo.3451524
31. Michael, L., Miyuki, K., Jiro, G.: Japanese Female Facial Expression (JAFFE) Database, figshare. J. Contrib https://doi.org/10.6084/m9.figshare.5245003.v2
32. Zhang, K., Zhang, Z., Li, Z., Qiao, Y.: Joint Face Detection and alignment using multi-task cascaded convolution networks. IEEE Signal Process. Lett. **23**(10), 1499–1503 (2016)
33. Aneesh, K.U., Hasan, A.Z., Pai, S.N.: Facial emotion recognition-a gift for the visionless. Int. J. Sci. Technol. Res. **9**(6), 228–233 (2020)
34. Wang, Z., Sun, X.: Manifold adaptive kernel local fisher discriminant analysis for face recognition. J. Multim. **7** (2012)
35. Arriaga, O., Paul, G.P, Matias, V.: Real-time Convolutional Neural Networks for Emotion and Gender Classification, pp 1–5 (2017)
36. Eidinger, E., Enbar, R., Hassner, T.: Age and Gender Estimation of Unfiltered Faces in IEEE. Trans. Inform. Forensi. Sec. **9**(12), 2170–2179 (2014)